

BHUVANESWAR SARAKANAM

AI/ML Engineer | Generative AI · LLMs & Agents · RAG · MLOps

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SUMMARY

AI/ML engineer with 6+ years designing, building, and deploying production generative-AI and machine-learning systems in regulated finance and healthcare. Specializes in LLM applications and agentic AI — multi-agent orchestration, tool-calling, RAG, and MCP — backed by strong Python engineering and end-to-end MLOps across AWS, Azure, and GCP. Proven record taking AI from prototype to production: reducing query resolution time 30%, cutting manual review 60% through multi-agent automation, and delivering predictive models that improved forecasts 18% and saved ~\$1.2M annually. Master's in Computational Data Science (Purdue University).

TECHNICAL SKILLS

- **Generative & Agentic AI:** LLMs (GPT, Claude, Llama, Mistral), RAG architectures, agentic frameworks (LangChain, LangGraph, CrewAI, LlamaIndex), tool-calling, Model Context Protocol (MCP), prompt engineering, fine-tuning & evaluation, vector databases (Pinecone, FAISS, Chroma, pgvector), vLLM inference optimization
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, deep learning, NLP (Hugging Face Transformers), forecasting, clustering, anomaly detection, graph ML (GraphSAGE), model evaluation, explainability (SHAP, LIME)
- **Programming:** Python (Pandas, NumPy, FastAPI, SQLAlchemy), SQL (CTEs, window functions), Java, Bash; REST APIs, microservices, object-oriented design
- **MLOps & Deployment:** MLflow, Docker, Kubernetes (EKS/AKS), CI/CD (GitHub Actions, Jenkins, Azure DevOps), model serving & monitoring, drift detection, observability (Prometheus, Grafana), AI-assisted development (GitHub Copilot, Cursor)
- **Cloud & Data:** AWS (SageMaker, Bedrock, Lambda, S3, EC2, CloudWatch), Azure (ML Studio, OpenAI, Data Factory), GCP (Vertex AI, BigQuery), Snowflake, PostgreSQL; ETL/ELT (Apache Airflow, Kafka), data pipelines
- **Collaboration:** Cross-functional delivery with product, engineering, and data science; stakeholder communication; team leadership & mentoring; Agile (Scrum, JIRA)

PROFESSIONAL EXPERIENCE

AI/ML Engineer, New York Life Insurance

10/2024 – Present | Remote, USA

- Designed and deployed domain-specific LLM and RAG systems for policy advisory and customer engagement, cutting query resolution time 30% and improving retrieval accuracy 22% via semantic vector search (Pinecone, FAISS).
- Architected multi-agent workflows (LangChain, LangGraph) with tool-calling and MCP integrations, automating underwriting and claims processing with human-in-the-loop review.
- Led development of churn and lifetime-value models improving customer-lifetime-value forecasts 18%, partnering with actuarial and marketing teams.
- Built end-to-end MLOps pipelines on AWS (SageMaker, Lambda) with MLflow, Docker, and CI/CD — reproducible experimentation, automated evaluation, and drift-triggered retraining.
- Delivered executive dashboards (Power BI on Snowflake); implemented observability (Prometheus, Grafana) cutting debugging time 25%; mentored junior engineers on explainability and MLOps standards.

Machine Learning Engineer, Tata Consultancy Services

10/2021 – 06/2023 | Hyderabad, India

- Led a 5-member team productionizing healthcare NLP and RAG systems on Azure ML and Kubernetes, cutting deployment cycles 40%.
- Developed NLP entity-recognition and classification models (Hugging Face, TensorFlow) reaching 94% accuracy; owned evaluation methodology and error analysis.
- Built RAG document-intelligence solutions (Azure Cognitive Search + LangChain) accelerating clinical retrieval 35%; methodology whitepaper adopted across 3 client accounts.

- Developed backend services and APIs (Python, Java) and CI/CD pipelines (Azure DevOps), shortening delivery cycles 25%.

Data Scientist, Omega Healthcare Management Services 01/2019 – 09/2021 | Chennai, India

- Built claim-denial and readmission-risk models (Scikit-learn, XGBoost) on messy real-world data, tuning precision/recall thresholds — reduced reprocessing 22%, saving ~\$1.2M annually.
- Developed a GraphSAGE graph neural network for real-time fraud and anomaly detection over relational transaction data.
- Engineered real-time streaming pipelines (Kafka, PostgreSQL, Airflow) processing 50K+ records daily for training and inference.
- Built NLP models for medical text summarization and ICD code extraction; conducted fairness and bias evaluations; deployed via Docker and Kubernetes.

PROJECTS

Agentic Workflow Automation Platform (LangGraph + CrewAI + MCP)

- Multi-agent system with tool-calling agents (SQL, API lookups, document processing), MCP servers, and human-in-the-loop review — reduced manual review time 60%.

LLM-Powered RAG Search Assistant (vLLM)

- Document ingestion, chunking, and embedding pipelines (LangChain, OpenAI embeddings, FAISS) with re-ranking and conversational memory; vLLM optimization cut token latency 40%.

End-to-End MLOps Pipeline (AWS + MLflow)

- Full lifecycle automation — data, training, deployment, monitoring — with MLflow Registry, Docker, SageMaker, and drift triggers (Lambda + CloudWatch).

Credit-Card Fraud Detection (GraphSAGE GNN)

- Graph neural network detecting coordinated fraud in transaction networks with real-time, precision-focused risk scoring.

EDUCATION

Master of Science, Computational Data Science — Purdue University, West Lafayette, IN	<i>08/2023 – 05/2025</i>
B.Tech, Electronics and Communication Engineering — Adikavi Nannaya University, India	<i>06/2017 – 05/2021</i>